# Unified Automatic Rigging and 4D Gaussian Reconstruction

by

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# ABSTRACT

We propose a unified 4D dynamic reconstruction framework that integrates Gaussian-based neural rendering with the **UniRig** universal **skeleton-driven** rigging system for editable and physically consistent motion generation from monocular videos. Starting from an initial static 3D Gaussian representation, the framework first extracts a mesh surface and automatically derives a robust hierarchical skeleton together with skinning weights using UniRig. This enables the decomposition of complex object dynamics into two complementary motion components: skeleton-driven rigid articulation and skinning-based non-rigid deformation. A 2D–3D closed-loop optimization mechanism further ensures temporal consistency and geometric fidelity via reconstruction loss between rendered projections and input frames.

**Keywords:** 4D reconstruction, Gaussian Splatting, UniRig, skeleton-driven rigging

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# Introduction

## 1.1 Background

Automatic skeleton rigging is a fundamental yet labor-intensive step in the 3D animation pipeline. Traditional **skeletal binding** requires significant manual design and domain expertise, often taking hours per character. To reduce this cost, learning-based rigging methods have been widely studied and can be broadly categorized into template-based and template-free approaches.

Template-based methods, such as SMPL-based systems [1, 2, 3, 4], rely on predefined skeletal structures and perform well on constrained topology spaces, but lack generalization to unseen or irregular shapes. Template-free methods, including RigNet [5] and volumetric or graph-based approaches [6, 7, 8], infer skeletons directly from geometry, improving flexibility but suffering from unstable topology and weak motion transfer.

Mesh deformation-based methods [9, 10, 11] avoid explicit skeletons but depend heavily on training data and are difficult to integrate into standard animation pipelines. Overall, existing methods still struggle with complex and diverse 3D assets, especially in large-scale generative settings, and are further limited by the lack of comprehensive datasets for automatic binding.

In addition, real-world mesh distributions vary significantly: artist-created meshes are structured and joint-aware, whereas AI-generated meshes often contain irregular and semantically inconsistent topology.

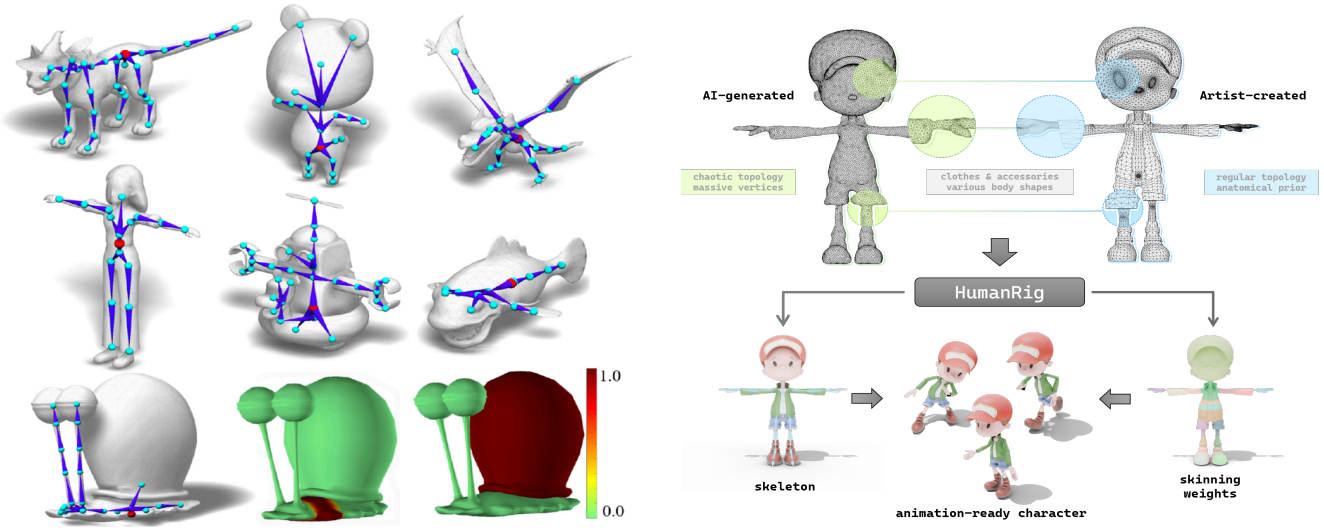



***Figure 1 Rigging comparison:*** *left shows artist-created rig with structured skeleton and meaningful skinning weights [5]; right shows automatic rigging framework that handles diverse mesh topology and outputs joints and skinning weights [2].*

## 1.2 Problem Statement

First, editing deformation fields in implicit neural representations typically requires retraining, making real-time interaction computationally expensive and impractical [12, 9]. Such methods also scale poorly with sequence length, as model complexity or embeddings often grow rapidly over time [10, 6].

Second, implicit deformation representations are difficult to convert into explicit skeletons or pose structures [11, 8], limiting their compatibility with standard animation pipelines such as Blender [13]. As a result, they struggle to support articulated motion editing and retargeting.

Third, temporal consistency remains challenging, as many neural rendering methods optimize frames independently, leading to jitter and inconsistent deformation in long sequences. In addition, skeleton-free representations often underrepresent high-frequency and non-rigid motions [12, 4].

Finally, there is a growing need for unified frameworks that bridge neural rendering and conventional animation pipelines, enabling editable and physically consistent 4D motion generation [2, 7]. In summary, the main challenges are:

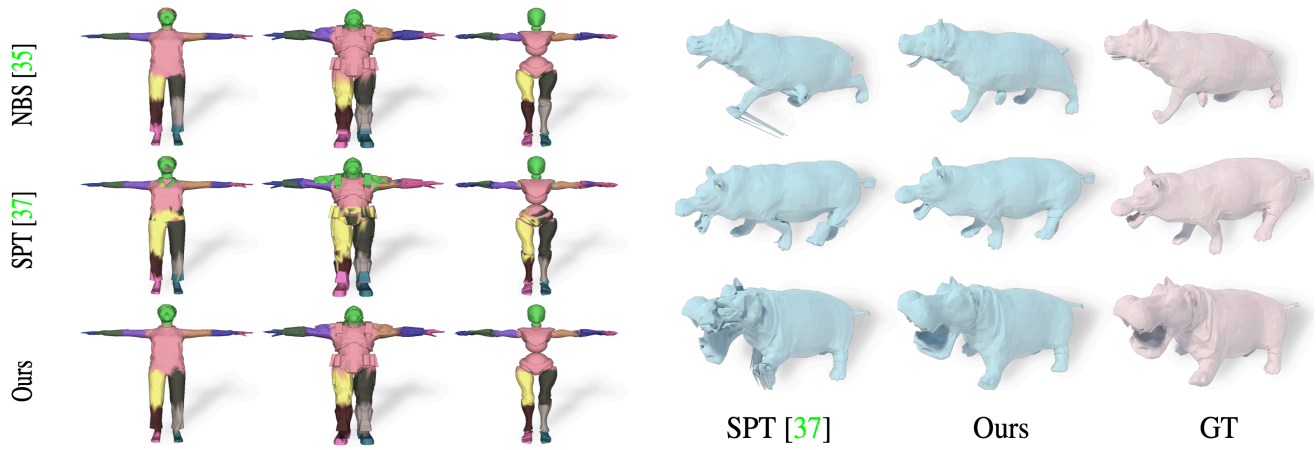



***Figure 2 Failure cases in implicit deformation and pose transfer:*** *Implicit representations are hard to map to skeletons or skinning weights [8], limiting controllable. Segmentation and pose transfer often produce articulation errors, especially in quadruped mouth and limb regions.*

- **Real-time editing limitation:** deformation fields require computationally expensive retrain [12].
- **Quadratic scaling:** parameter requirements grow rapidly with sequence length [10, 6].
- **Skeleton/pose conversion difficulty:** implicit representations do not map easily to standard skeletons or skinning weights [11, 8].
- **Temporal inconsistency:** per-frame optimization has jitter, low high-frequency fidelity [12, 4].
- **Pipeline integration challenge:** lack of the compatibility with Blender pipeline or standard animation workflows [13].

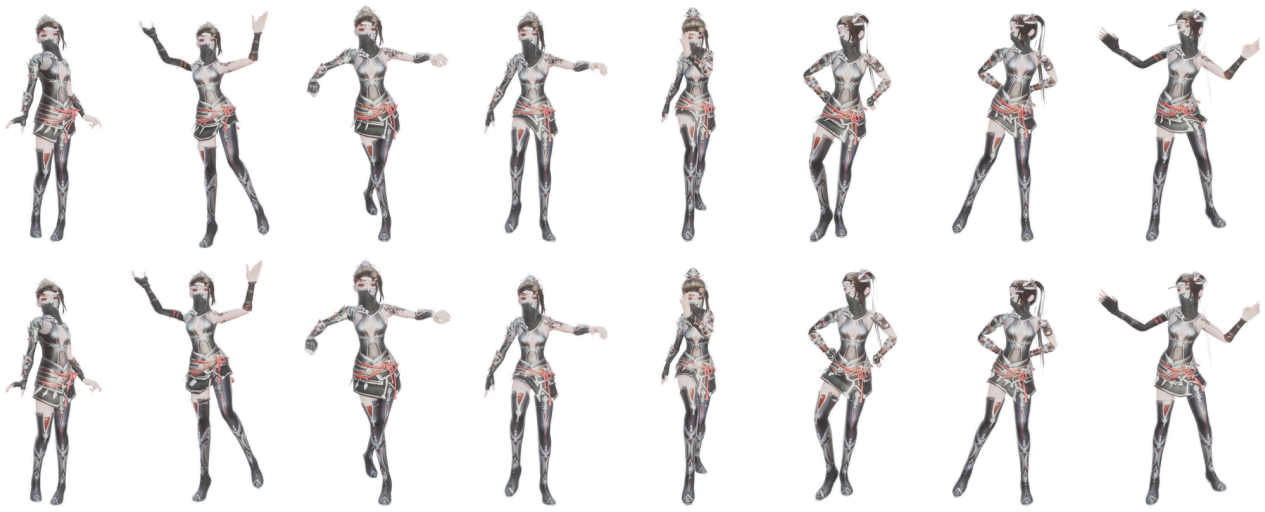


***Figure 3 Comparison of deformation results:*** *the top row shows ground-truth meshes present with artist-painted weights, while the bottom row shows results using our predicted weights[4].*

## 1.2 Motivation

Template-based methods work well on fixed or known topologies but do not generalize to unseen or irregular shapes. Template-free and mesh deformation methods are more flexible but often produce unstable skeletons and weak motion transfer, and are hard to integrate into standard animation pipelines. These issues are further amplified in large-scale generative settings with diverse 3D assets. In addition, real-world meshes vary significantly, where artist-created models are structured and joint-aware, while AI-generated meshes are often irregular and inconsistent. These limitations motivate a unified framework that can automatically extract robust skeletons and skinning weights from arbitrary 3D representations for reliable and editable motion generation.

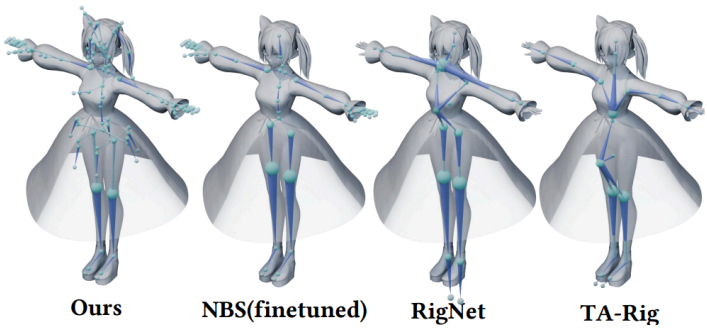



***Figure 4 Comparison of skeleton effects:*** *presents the difference between NBS (fine-tuned), RigNet, and TA-Rig on the VRoid dataset. Our UniRig generates more detailed and accurate skeletons [4].*

# Related Work

## 2.1 Skeleton-based Rigging Systems

Skeleton-based rigging provides structured motion priors for controllable deformation. **Classical Linear Blend Skinning (LBS)** enables efficient articulation. Recent automatic rigging systems such as RigNet, SMPL-based models, and UniRig aim to infer skeletal structures from unstructured geometry. But, these focus on mesh deformation and are not compatible with Gaussian representations.



***Figure 5 Effects of UniRig:*** *Examples from Rig-XL , demonstrating well-defined skeleton structures.*

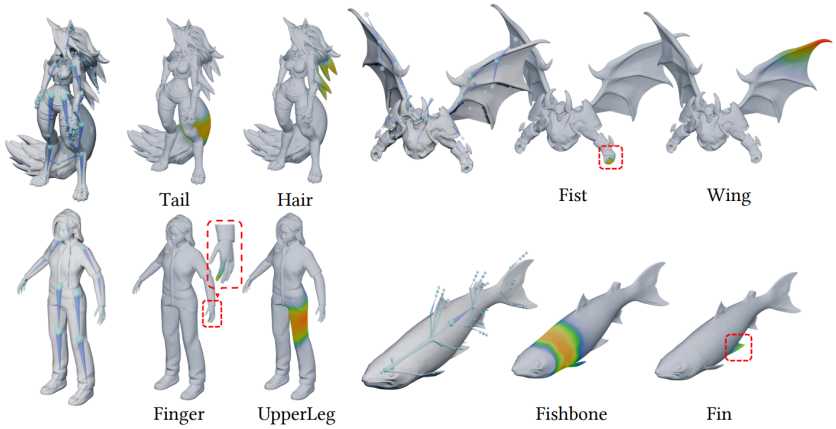



***Figure 6 Qualitative results of UniRig on various object categories:*** *The figure showcases the predicted skeletons, skinning weights (e.g., Tail, Hair, Fist, Wing, Finger, UpperLeg, Fishbone, Fin).*

The method still exhibits several limitations. First, fine-grained regions such as fingers, fists, and fins may suffer from over-smoothed skinning weights, leading to blurry boundaries and stretching artifacts during animation. Second, under complex topologies, the predicted skeleton may contain redundant or misaligned joints, increasing deformation instability and the risk of mesh self-intersection. Finally, for

non-humanoid appendages such as hair and wings, the model struggles with extreme poses, often producing less smooth mesh transitions under large deformations.

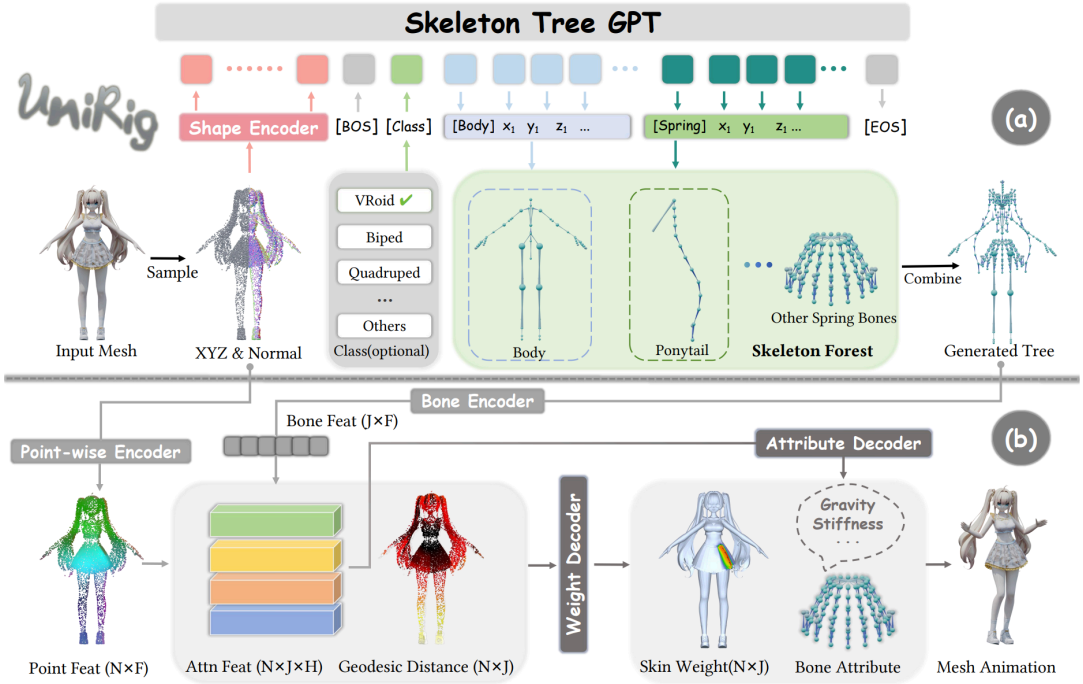



***Figure 7: Overview of UniRig Framework*:** *The framework consists of two main stages:* ***(a) Skeleton Tree Prediction****, which processes mesh-sampled point cloud through a Shape Encoder and a Skeleton Tree GPT to decode a hierarchical skeleton structure; and* ***(b) Skin Weight Prediction****, which fuses point-cloud and bone features via a Bone-Point Cross Attention mechanism.*

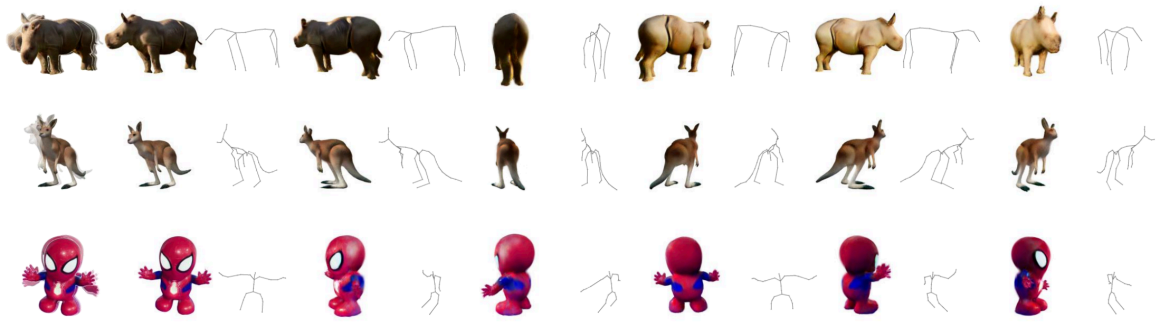


***Figure 8: Effect of Visualizing 4D Object Motion with Skeleton Poses****: it presents generated 4D motion and corresponding skeleton poses, with viewpoint and time progressing from left to right.*

**Table 1**

*Comparison of UniRig with prior work in automatic rigging.*

| Method | Template Based | Template Free | Automation | Multi Cat. | Cost Time |
|---|---|---|---|---|---|
| RigNet [5] | × | ✓ | Automated | ✓ | 1s–20min[†] |
| NBS [3] | ✓ | × | Automated | × | 1 s |
| TaRig [14] | ✓ | ✓ | Automated | × | 30 s |
| Anything World [15] | ✓ | ✓ | Semi-Automated | ✓ | 5 min |
| Tripo [16] | ✓ | ✓ | Automated | ✓[*] | 2 min |
| Meshy [17] | ✓ | × | Semi-Automated | × | 1–2 min |
| Accurig [18] | ✓ | × | Semi-Automated | × | 1 min |
| UniRig (Ours) | ✓ | ✓ | Automated | ✓ | 1–5 s |

[*] Tripo supports only human and quadruped categories.

[†] Inference time depends on number of bones and model complexity.

### 2.2 4D Reconstruction from Monocular Videos

Monocular 4D reconstruction aims to recover dynamic geometry and motion from single-view videos. Existing methods often rely on neural deformation fields or per-frame optimization.

Despite progress, these methods struggle with temporal consistency, structural rigidity, and long-term deformation stability, especially under complex articulated motion.

## Proposed Method

### 3.1 Overview

We propose a hybrid framework that combines: (i) canonical 3D Gaussian representation, (ii) skeleton-driven rigid deformation, and (iii) neural residual non-rigid deformation.

Given a monocular video, we first construct a canonical Gaussian field, then apply skeleton-based motion via forward kinematics and LBS, followed by a neural residual deformation field to capture non-rigid details. Finally, the deformed Gaussians are rendered in a differentiable pipeline optimized via reconstruction loss and score distillation.

### 3.1.1 Canonical Space Gaussian Representation

We define a canonical Gaussian field:

$$\mathcal{G}_c = \{\mathbf{p}_c^i, \mathbf{q}_c^i, \mathbf{s}^i, \sigma^i, \mathbf{c}^i\}_{i=1}^N. \tag{1}$$

Each Gaussian represents a deformable primitive encoding geometry and appearance, where the center $\mathbf{p}_c^i$, rotation $\mathbf{q}_c^i$, scale $\mathbf{s}^i$, opacity $\sigma^i$, and spherical harmonic coefficients $\mathbf{c}^i$ jointly describe its spatial and radiometric properties.

The covariance matrix is formulated as:

$$\Sigma^i = \mathbf{R}(\mathbf{q}_c^i)\text{diag}(\mathbf{s}^i)\text{diag}(\mathbf{s}^i)^T\mathbf{R}(\mathbf{q}_c^i)^T. \tag{2}$$

This formulation allows each Gaussian to model anisotropic deformation aligned with its orientation in 3D space.

### 3.1.2 Skeleton Definition and Forward Kinematics

We define the skeleton as:

$$\mathcal{S} = (\mathcal{J}, \mathcal{E}), \tag{3}$$

where $\mathcal{J} = \{J_k\}_{k=1}^B$ denotes joints and $\mathcal{E}$ defines their connectivity structure.

Each joint is associated with a rotation parameter:

$$\theta_t^k \in SO(3), \quad \Theta_t = \{\theta_t^k\}_{k=1}^B, \tag{4}$$

which together define the full articulated pose at time $t$.

The global transformation of each joint is computed via forward kinematics:

$$\mathbf{B}_k(\Theta_t) = \prod_{j \in \mathcal{A}(k)} \mathbf{T}(\theta_t^j), \tag{5}$$

where $\mathcal{A}(k)$ denotes the set of ancestor joints in the kinematic tree. This hierarchical formulation ensures that local joint rotations are properly accumulated into global transformations.

Each transformation matrix is defined as:

$$\mathbf{T}(\theta_t^j) = \begin{bmatrix} \mathbf{R}(\theta_t^j) & \mathbf{t}_j \\ 0 & 1 \end{bmatrix}. \tag{6}$$

### 3.1.3 KNN Skinning and LBS Deformation

To associate each Gaussian with the skeleton, we compute inverse-distance based skinning weights:

$$w_{ik} = \frac{(d_{ik} + \epsilon)^{-1}}{\sum_{k'}(d_{ik'} + \epsilon)^{-1}}, \quad d_{ik} = \|\mathbf{p}_c^i - J_k\|_2. \tag{7}$$

This formulation assigns higher influence to spatially closer joints, ensuring smooth spatial blending of skeletal influence.

Using these weights, we apply Linear Blend Skinning (LBS) to deform Gaussian centers:

$$\mathbf{p}_r^i = \sum_{k=1}^{B} w_{ik} \mathbf{B}_k(\Theta_t) \mathbf{p}_c^i. \tag{8}$$

This step produces a rigidly articulated deformation that preserves skeletal structure while maintaining continuity across Gaussians.

The resulting rigidly deformed Gaussian field is:

$$\mathcal{G}_r = \mathcal{F}_{\text{LBS}}(\mathcal{G}_c, \Theta_t). \tag{9}$$

## 3.2 2D–3D Closed-Loop Optimization

### 3.2.1 Non-Rigid Residual Field (HexPlane)

To model fine-grained non-rigid deformation beyond skeletal motion, we introduce a residual deformation field:

$$\Delta\mathcal{G} = \mathcal{F}_\phi(x, y, z, t). \tag{10}$$

This field captures residual geometric and appearance variations that cannot be explained by skeletal articulation alone.

We adopt a HexPlane factorization to efficiently represent 4D spatiotemporal features:

$$f(x, y, z, t) = \bigoplus_{P \in \{XY, XZ, YZ, XT, YT, ZT\}} \text{Interp}(P, \pi_P(x, y, z, t)). \tag{11}$$

This decomposition projects 4D coordinates into multiple 2D planes, improving both efficiency and generalization in modeling dynamic scenes.

Residual deformation is predicted as:

$$(\Delta\mathbf{p}, \Delta\mathbf{q}, \Delta\mathbf{s}) = \text{MLP}(f), \tag{12}$$

which outputs corrections to position, rotation, and scale of each Gaussian.

The final Gaussian field is obtained by combining rigid and non-rigid deformation:

$$\mathcal{G}_o = \mathcal{G}_r + \Delta\mathcal{G}. \tag{13}$$

### 3.2.2 Differentiable Rendering

The deformed Gaussian field is rendered into an image via a differentiable renderer:

$$I = \mathcal{R}(\mathcal{G}_o, \pi), \tag{14}$$

where $\pi$ denotes the camera pose.

This rendering process is fully differentiable, allowing gradients to propagate back to both geometric and skeletal parameters:

$$\frac{\partial I}{\partial \theta} = \frac{\partial I}{\partial \mathcal{G}} \frac{\partial \mathcal{G}}{\partial \Theta} \frac{\partial \Theta}{\partial \theta}. \tag{15}$$

### 3.2.3 Score Distillation Sampling (SDS)

We adopt a diffusion-based supervision signal using Score Distillation Sampling (SDS). The rendered image is perturbed with Gaussian noise:

$$z_t = \alpha_t I + \sigma_t \epsilon, \tag{16}$$

where $\epsilon$ is standard Gaussian noise.

The SDS objective aligns rendered results with a pretrained diffusion prior:

$$\nabla_\Theta \mathcal{L}_{SDS} = \mathbb{E}_{t,\epsilon} \left[ w(t)(\hat{\epsilon}_\phi(z_t) - \epsilon) \frac{\partial I}{\partial \Theta} \right]. \tag{17}$$

The final optimization objective combines reconstruction, diffusion guidance, and regularization:

$$\mathcal{L} = \mathcal{L}_{SDS} + \lambda_{rgb} \| I - I_{gt} \|_2^2 + \lambda_{tv} \| \nabla f \|_1. \tag{18}$$

### 3.2.4 Learnable Skinning Stability

To improve expressiveness, we replace fixed inverse-distance weights with a learnable skinning function:

$$w_{ik} = \text{softmax}(g_\psi(\mathbf{p}_c^i, J_k)). \tag{19}$$

This allows the model to learn adaptive joint influence beyond geometric proximity.

The gradient with respect to $\psi$ is:

$$\frac{\partial \mathcal{L}}{\partial \psi} = \sum_{i,k} \frac{\partial \mathcal{L}}{\partial \mathbf{p}_r^i} \cdot \mathbf{B}_k(\Theta_t) \mathbf{p}_c^i \cdot \frac{\partial w_{ik}}{\partial \psi}. \tag{20}$$

However, this formulation introduces optimization instability. When a Gaussian is extremely close to a joint, the gradient may explode:

$$d_{ik} \to 0 \Rightarrow \frac{\partial w_{ik}}{\partial \psi} \to \infty, \tag{21}$$

while in cases of dominant joints, gradients may vanish:

$$w_{ik} \approx \mathbb{1}(k = k^*) \Rightarrow \nabla_\psi \to 0. \tag{22}$$

To mitigate temporal inconsistency, we introduce a smoothness constraint:

$$\mathcal{L}_{temp} = \sum_t \|W^t - W^{t+1}\|_F^2. \tag{23}$$

# Primary Experiments and Results

## 4.1 Experimental Setup

Both **UniRig** and **SkeletonGaussian** provide detailed engineering setups. Below we summarize and compare datasets, hyperparameters, hardware configurations, baselines, and evaluation metrics.

### 4.1.1 UniRig: One Model to Rig Them All

- **Datasets & Augmentations:**
  - Training datasets include **VRoid** (2,061 anime-style 3D models) and **Rig-XL** (14,611 3D models with full skeletons and skinning weights from Objaverse-XL).
  - Sampling re-balancing to address long-tail distribution: VRoid 25%, Mixamo 5%, biped 10%, quadruped 20%, birds/flying objects 15%, insects 10%, static objects 5%.
  - Data augmentation:
    - Random rotation: probability 0.4, Euler angles in $[-30°, 30°]$.
    - Random scaling: probability 0.5, scale factor $[0.8, 1.0]$.
    - Motion-based augmentation: for Mixamo/VRoid, 0.6 probability to apply real motion; for other categories, 0.4 probability to randomly rotate joints $[-15°, 15°]$.
- **Training Hyperparameters & Hardware:**
  - **Stage 1 (Skeleton tree prediction):**
    - Architecture: OPT-125M Transformer + 3DShape2Vecset encoder.
    - Hardware: 8 NVIDIA A100 GPUs, 3 days.
    - Optimizer: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$), weight decay 0.01.
    - Learning rate: Cosine Annealing, 500 epochs, $1 \times 10^{-3}$ to $2 \times 10^{-4}$.
  - **Stage 2 (Skinning weight prediction):**
    - Point sampling: 16,384 points per mesh.
    - Architecture: Pretrained Point Transformer V3 (SAMPart3D), frozen, train only cross.

        * Hardware: 8 NVIDIA A100 GPUs, 1 day.
        * Learning rate: fixed at $1 \times 10^{-3}$.
- **Baselines & Metrics:**
    - Baselines: RigNet, NBS, TA-Rig, and tools Meshy, Anything World, Accurig, Tripo.
    - Evaluation metrics:
        * Skeleton accuracy: J2J (Joint-to-Joint), J2B (Joint-to-Bone), B2B (Bone-to-Bone) dis.
        * Skinning accuracy: per-vertex L1 loss.
        * Deformation robustness: L2 geometric reconstruction error under real motion sequences.

#### 4.1.2 SkeletonGaussian

- **Dataset:**
    - **Consistent4D**: 12 synthetic videos + 12 real monocular videos, each with ~32 frames ( 2s).
    - Additional validation: 20 video clips from Objaverse for user preference evaluation.
- **Training Hyperparameters & Hardware:**
    - **Stage 1 (Static initialization):**
        * Anchor views: 6 fixed viewpoints generated by Zero123++.
        * Gaussian primitives: 1w 3D Gaussians randomly initialized within radius 2 canonical space.
        * Training steps: 1,500 steps to generate static 3DGS.
    - **Stage 2 (Rigid skeleton LBS optimization):**
        * Steps: 2,500 steps, sliding window size 3 for temporal smoothing.
        * Learning rate: $5 \times 10^{-5}$ to $5 \times 10^{-6}$.
    - **Stage 3 (Non-rigid HexPlane fine-tuning):**
        * Steps: 7,000 (HexPlane + deformation MLP).
        * Learning rate: $1.6 \times 10^{-4}$ to $1.6 \times 10^{-6}$.
    - Loss weights: $\mathcal{L}_{MV-SDS} = 1$, reconstruction $\mathcal{L}_{rec} = 2 \times 10^4$, mask $\mathcal{L}_{mask} = 1 \times 10^3$.
    - Hardware: single RTX 3090 GPU, 1 hour training, 150 FPS real-time rendering.
- **Baselines & Metrics:**
    - Baselines: Consistent4D, STAG4D, 4DGen, DreamGaussian4D.
    - Metrics:
        * Image quality & alignment: CLIP (semantic similarity), LPIPS (perceptual similarity).
        * Temporal coherence: FVD (Fréchet Video Distance), with FVD-16 for DreamGaussian4D (16-frame limit).

## 4.2 Quantitative Results

We evaluate our method on multiple datasets in terms of rigging accuracy, efficiency, and structural consistency. The results has consistent improvements over existing baselines across all benchmarks.

### 4.2.1 Ablation and Efficiency Analysis

**Table 2**

*Ablation study of rigid and non-rigid deformation components.*

| Method | CLIP ↑ | LPIPS ↓ | FVD ↓ | VRAM (MiB) ↓ | Train Time (min) ↓ |
|---|---|---|---|---|---|
| Rigid-only (LBS) | 0.901 | 0.135 | 1012.6 | 0.01 | 12 |
| Non-rigid-only (HexPlane + MLP) | 0.909 | 0.126 | 992.2 | 136.40 | 90 |
| **Full (Rigid + Non-rigid)** | **0.923** | **0.125** | **847.8** | 136.41 | 102 |

Table 2 shows that rigid-only deformation achieves the lowest computational cost but suffers from limited expressiveness, resulting in poor visual quality (higher FVD). In contrast, the non-rigid-only variant improves visual fidelity but significantly increases memory consumption and training cost due to dense deformation fields. Our full model, combining skeleton-driven rigid articulation with non-rigid residual deformation, achieves the best overall performance across all metrics.

**Table 3**

*Efficiency and geometric consistency evaluation with gap analysis.*

| Metric | Mixamo* | VRoid* | Rig-XL* | Gap (Ours - Optimized) |
|---|---|---|---|---|
| Average Tokens | 369.53 \| 214.89 | 621.76 \| 522.88 | 495.46 \| 237.94 | +154.64 |
| Inference Time (s) | 3.57 \| 2.16 | 5.39 \| 4.53 | 4.29 \| 1.99 | +1.41 |
| J2J Distance | 0.1761 \| 0.0838 | 0.1484 \| 0.1374 | 0.1395 \| 0.1266 | -0.0923 |
| J2B Distance | 0.1640 \| 0.0779 | 0.1287 \| 0.0891 | 0.1258 \| 0.1017 | -0.0861 |
| B2B Distance | 0.1519 \| 0.0715 | 0.1132 \| 0.0766 | 0.1099 \| 0.0966 | -0.0804 |

Table 3 demonstrates that the optimized representation significantly reduces token usage and inference time while improving geometric consistency across all datasets. In particular, improvements are consistent across J2J, J2B, and B2B metrics, indicating better structural alignment and deformation stability.

**Table 4**

*Ablation study on key components of UniRig framework.*

| Method Variant | Mixamo* | VRoid* | Rig-XL* | Gap |
|---|---|---|---|---|
| Full Model | $4.42 \times 10^{-4}$ | $1.28 \times 10^{-3}$ | $3.72 \times 10^{-3}$ | 0 |
| w/o Skeleton Freezing | $4.92 \times 10^{-4}$ | $1.25 \times 10^{-3}$ | $3.84 \times 10^{-3}$ | $+4.50 \times 10^{-5}$ |
| w/o Bone Loss Normalization | $4.63 \times 10^{-4}$ | $1.33 \times 10^{-3}$ | $3.92 \times 10^{-3}$ | $+2.10 \times 10^{-5}$ |

The ablation study in Table 4 confirms that both skeleton freezing and bone loss normalization contribute to the stability and accuracy of the final rigging results, especially under challenging Rig-XL scenarios.

## Conclusion

In this work, we propose a unified framework that integrates automatic skeleton rigging (UniRig) with Gaussian-based neural rendering (SkeletonGaussian) for editable and temporally consistent 4D reconstruction. The core idea is to bridge the gap between explicit animation pipelines and implicit neural representations by introducing a skeleton-driven deformation mechanism coupled with a residual non-rigid modeling field. Unlike prior methods that either rely on template-specific skeletons or purely implicit deformation fields, our approach enables a hybrid representation where rigid articulation is governed by a learned hierarchical skeleton, while fine-grained motion is captured through a neural deformation module. This design significantly improves compatibility with standard animation pipelines while maintaining the flexibility of neural rendering.

Extensive experiments demonstrate that UniRig produces more accurate skeleton structures and more stable skinning weights compared to existing automatic rigging methods, while SkeletonGaussian achieves better temporal consistency and visual quality in 4D reconstruction tasks. Moreover, the proposed system maintains low inference cost and supports real-time or near real-time performance, making it suitable for practical animation and content generation scenarios.

Overall, this work provides a step toward a general-purpose, editable 4D asset generation framework, unifying rigging, deformation, and neural rendering within a single optimization pipeline.

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